

Coupled Dispersion of a Two-Flexible-Plate Structural-Acoustic Waveguide with Low-Mach Poiseuille Flow

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Abstract

The coupled dispersion relation for a fluid column bounded by two flexible plates is derived in closed form and shown to factorise, for identical plates, into independent symmetric and antisymmetric families, recovering the single fluid-loaded-plate relation as a limit. A complete first-order asymptotic catalogue of the coupled wavenumbers is then assembled in the fluid-loading parameter and checked against exact numerical roots. A low-Mach Poiseuille mean flow is introduced through the Pridmore–Brown equation with plate-admittance boundary conditions; a Fredholm solvability argument gives a closed-form $O(M)$ correction to every branch, a critical layer appears at $M_{\text{crit}} = \Omega/\xi$, and a viscous analysis shows the discrete wall modes are not absorbed by it. Every analytical result is confirmed by three independent numerical methods.

Keywords: structural acoustics; fluid–structure coupling; dispersion relation; asymptotic analysis; Pridmore–Brown equation; critical layer

I Introduction

A duct with compliant walls supports waves that are neither purely structural nor purely acoustic but a coupling of the two, and the physics only becomes legible once each coupled branch is traced back to the uncoupled wave from which it grows. Sarkar and Sonti [1] achieved that tracing for a single flexible plate over a rigid wall by expanding the coupled wavenumbers in the fluid-loading parameter $\varepsilon = \rho a/m$, the ratio of the mass of the fluid column to the mass per unit area of the plate. Setting $\varepsilon = 0$ recovers the uncoupled wavenumbers exactly, and the first-order term then follows each coupled branch continuously away from its uncoupled parent.

This report extends that construction in two ways. First, the geometry is generalized from one flexible plate to a fluid column bounded by *two* flexible plates, in general non-identical; the coupled dispersion relation is derived in closed form, its exact symmetric and antisymmetric factorisation is established, and the single fluid-loaded-plate relation is recovered as a limit. Second, and more substantially, the quiescent column is allowed to carry a low-Mach Poiseuille mean flow through the Pridmore–Brown equation [3] with plate-admittance boundary conditions, a combination that does not appear to have been examined for this geometry. Table 1 lists the principal symbols.

II Coupled dispersion relation

Consider a two-dimensional acoustic fluid of density ρ and sound speed c occupying $0 \leq y \leq a$ and unbounded in x , with thin elastic plates closing the strip at $y = 0$ and $y = a$ and a time dependence $e^{i\omega t}$ understood. Defining the dimensionless admittance of each plate,

Table 1. Principal nomenclature.

Symbol	Meaning
$\varepsilon = \rho a/m$	fluid-loading parameter
α_j	dimensionless plate admittance
k_x, k_y	axial, transverse wavenumber
$\Omega = \omega/\omega_c$	non-dimensional frequency
$\xi = k_x/k_c$	non-dimensional axial wavenumber
$\tau = k_y a$	non-dimensional transverse wavenumber
M	centreline mean-flow Mach number
M_{crit}	critical-layer onset Mach number

sionless admittance of each plate,

$$\alpha_j \equiv \frac{k_y G_j}{\omega^2 \rho}, \quad j = 1, 2, \quad (1)$$

with $k_y = \sqrt{k^2 - k_x^2}$ and G_j the plate operator, the matched pressure and normal-velocity conditions give the coupled dispersion relation

$$(1 + i\alpha_1)(1 + i\alpha_2) = e^{2ik_y a} (1 - i\alpha_1)(1 - i\alpha_2), \quad (2)$$

which rearranges to the compact form

$$\tan(k_y a) = \frac{\alpha_1 + \alpha_2}{1 - \alpha_1 \alpha_2}. \quad (3)$$

Letting $\alpha_2 \rightarrow \infty$ (a rigid second wall) returns the classical single fluid-loaded-plate relation. For identical plates the relation factorises cleanly into independent symmetric and antisymmetric families,

$$(\text{SYM}) \quad \alpha = -\cot \frac{k_y a}{2}, \quad (\text{ANTI}) \quad \alpha = +\tan \frac{k_y a}{2}, \quad (4)$$

the symmetric branch being the single-plate problem with a replaced by $a/2$.

III Asymptotic catalogue

Non-dimensionalizing on the coincidence condition of plate 1, with $\Omega = \omega/\omega_c$, $\xi = k_x/k_c$, $l = k_c a$ and $\tau = l\sqrt{\Omega^2 - \xi^2}$, the two families take the compact form

$$(\text{SYM}) \quad \left(\frac{\xi^4}{\Omega^2} - 1 \right) \tau \tan \frac{\tau}{2} + \varepsilon = 0, \quad (5)$$

$$(\text{ANTI}) \quad \left(\frac{\xi^4}{\Omega^2} - 1 \right) \tau \cot \frac{\tau}{2} - \varepsilon = 0. \quad (6)$$

Setting $\varepsilon = 0$ exposes the uncoupled parents: the structural wave $\xi = \Omega^{1/2}$ (common to both families), the plane wave $\xi = \Omega$, and the cut-ons $\tau = m\pi$. Each parent is followed into

the coupled problem by writing it as the uncoupled value plus a correction. For the structural wave, with $\tau_s = l\sqrt{\Omega^2 - \Omega}$,

$$\xi_s^{\text{sym}} = \Omega^{1/2} - \frac{\varepsilon}{4l\sqrt{\Omega - 1} \tan(\tau_s/2)}, \quad (7)$$

$$\xi_s^{\text{anti}} = \Omega^{1/2} + \frac{\varepsilon \tan(\tau_s/2)}{4l\sqrt{\Omega - 1}}. \quad (8)$$

Where two parents approach one another the regular expansion fails and a local $\sqrt{\varepsilon}$ window opens a small gap between the branches. Every closed-form expression is checked against the exact numerical roots (Figs. 1, 3); for identical plates the symmetric and antisymmetric ladders are recovered cleanly (Fig. 2).

IV Mean flow and the critical layer

The column is now allowed to carry a Poiseuille mean flow $U(y) = U_0 f(\eta)$ with $f(\eta) = 4\eta(1 - \eta)$ and centre-line Mach number $M = U_0/c$. The pressure perturbation then satisfies the Pridmore–Brown equation, which in non-dimensional form reads

$$\frac{d}{d\eta} \left(\frac{1}{\hat{\sigma}^2} \frac{dp}{d\eta} \right) + \frac{\hat{\sigma}^2 - \xi^2}{\hat{\sigma}^2} p = 0, \quad (9)$$

with the Doppler-shifted frequency $\hat{\sigma} = \Omega - M\xi f$ and the plate admittance entering as a Robin condition at each wall. A Fredholm solvability argument applied to (9) gives a closed-form first-order correction to every coupled wavenumber of the quiescent catalogue,

$$\xi(M) = \xi^{(0)} + M\xi^{(1)} + M^2\xi^{(2)} + \dots, \quad (10)$$

in which the $O(M^2)$ term separates into distinct convection and shear contributions. A critical layer appears once the axial phase Mach number reaches unity, that is at

$$M_{\text{crit}} = \frac{\Omega}{\xi} \approx \sqrt{\Omega}, \quad (11)$$

and hence only for the subsonic structural wave below co-incidence. The coupled wavenumbers traced across the full nonlinear range of M , and the veering between the two structural waves for non-identical plates, are shown in Figs. 5, 6 and 7.

V Wall modes survive the critical layer

A viscous analysis of the layer (Figs. 8, 9) shows that the eigenvalue stays regular through M_{crit} and that the discrete wall modes are *not* absorbed by the critical layer: their decay is controlled by the wall Stokes layers, scales as $Re^{-1/2}$, and vanishes in the inviscid limit. The flow shifts the branches but does not destabilise them, so the system remains convectively stable across the validity region (Fig. 10). Every analytical result is corroborated by three independent numerical methods that agree to about one part in 10^9 .

VI Discussion

Taken on its own, the quiescent two-plate catalogue reorganizes for this geometry results consistent with the established quiescent literature [2, 5]; it is presented here as the setup that recovers and extends known results, not as the novelty. The genuinely new content is the combination with a sheared mean flow: the closed-form low-Mach corrections,

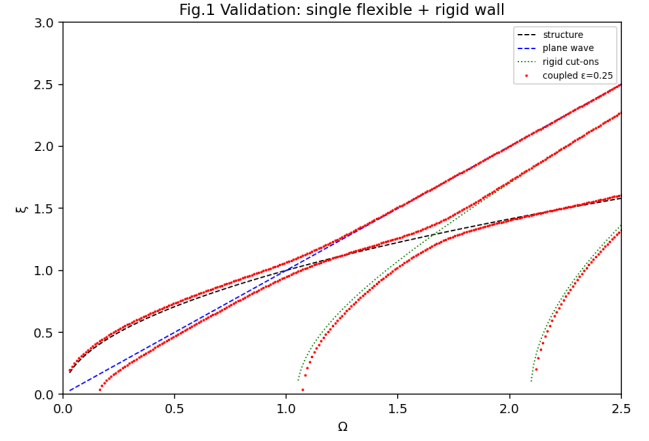


Figure 1. Closed-form first-order catalogue checked against the exact numerical roots.

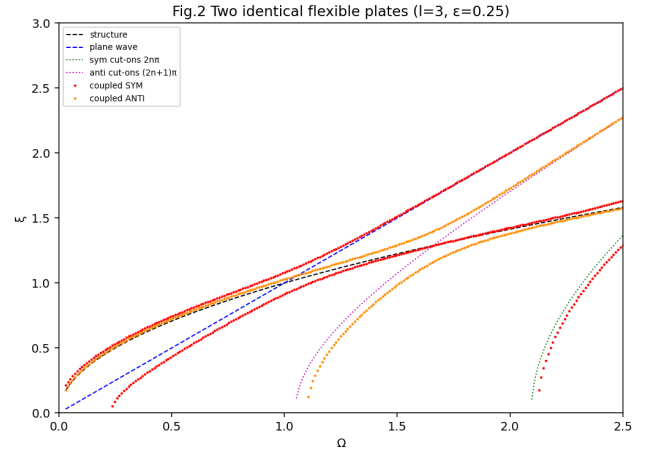


Figure 2. Symmetric and antisymmetric ladders for two identical plates.

the critical-layer threshold, and the result that the wall modes survive it. Framed this way the contribution is incremental but real, and suited to a journal such as the *Journal of Sound and Vibration* or *Wave Motion*.

References

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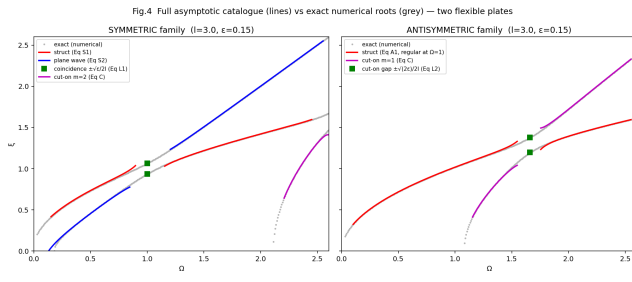


Figure 3. The full asymptotic catalogue of coupled wavenumbers for both symmetry families.

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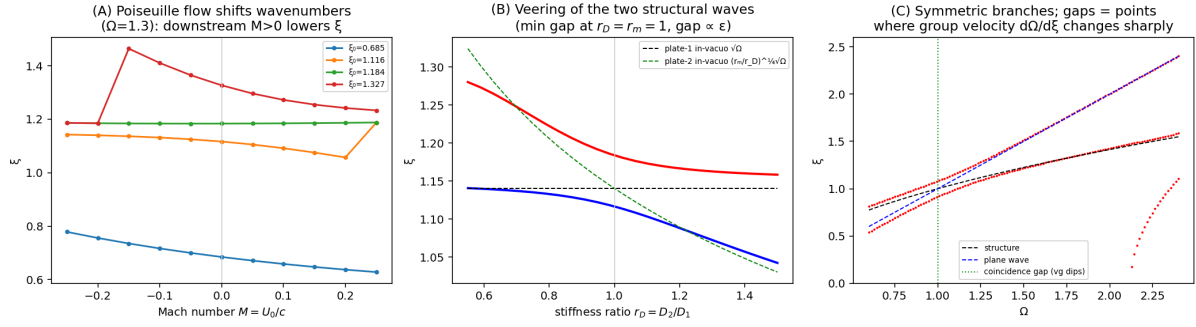


Figure 4. New mean-flow physics: the downstream wavenumber shift, the non-identical-plate veering, and the group-velocity reading of the gaps.

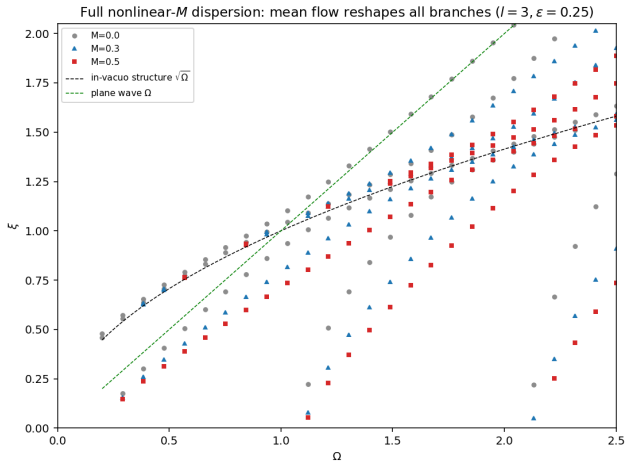


Figure 5. Coupled wavenumbers traced across the full nonlinear range of mean-flow Mach number.

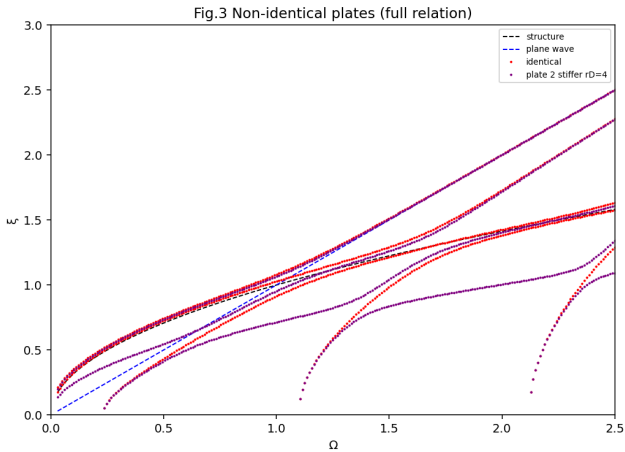


Figure 6. Veering between the two structural waves as the plates are made non-identical.

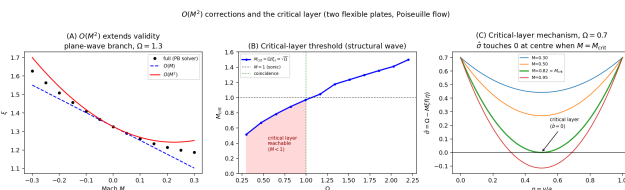


Figure 7. Behaviour near the critical condition, where the $O(M^2)$ convection and shear split matters.

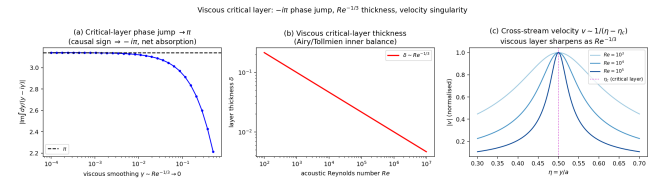


Figure 8. Viscous treatment of the critical layer; the wall modes are not absorbed and decay as $\text{Re}^{-1/2}$.

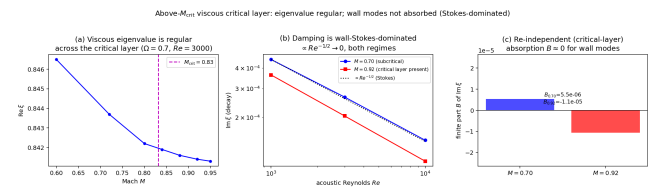


Figure 9. Global viscous spectrum across the parameter range.

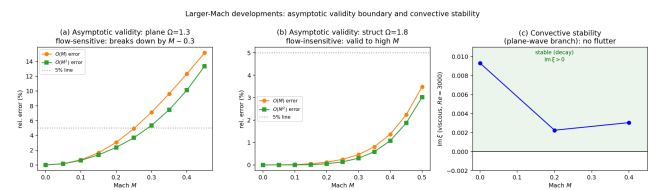


Figure 10. Validity and convective-stability map for the analysis.